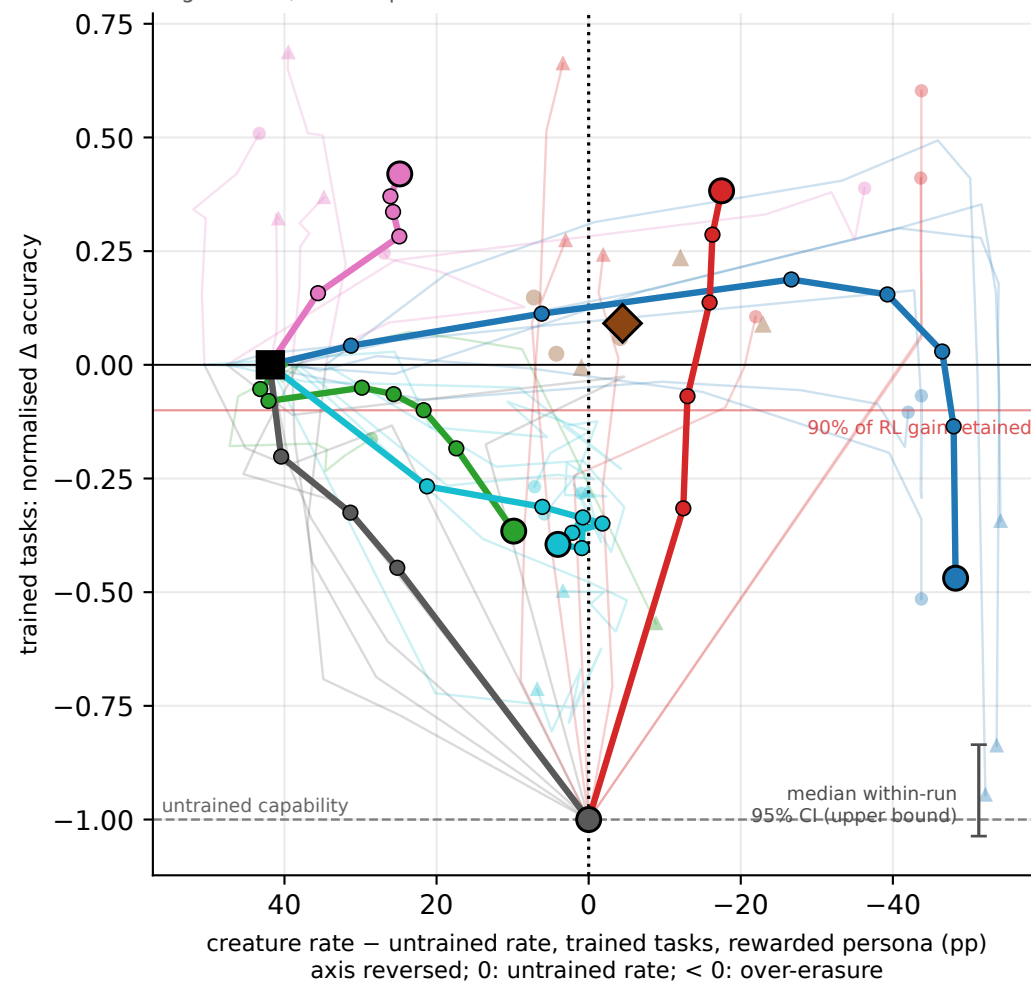


Dose-response of each repair method, all runs (Gemma s0, Gemma s1, Gemma s2, Qwen s0, Qwen s1, Qwen s3)
 Capability change is normalised by each run's RL gain on trained tasks (0: hacked anchor; -1: untrained capability on trained tasks)

A. Trained distribution

Optimal: $x = 0$ with maximal y

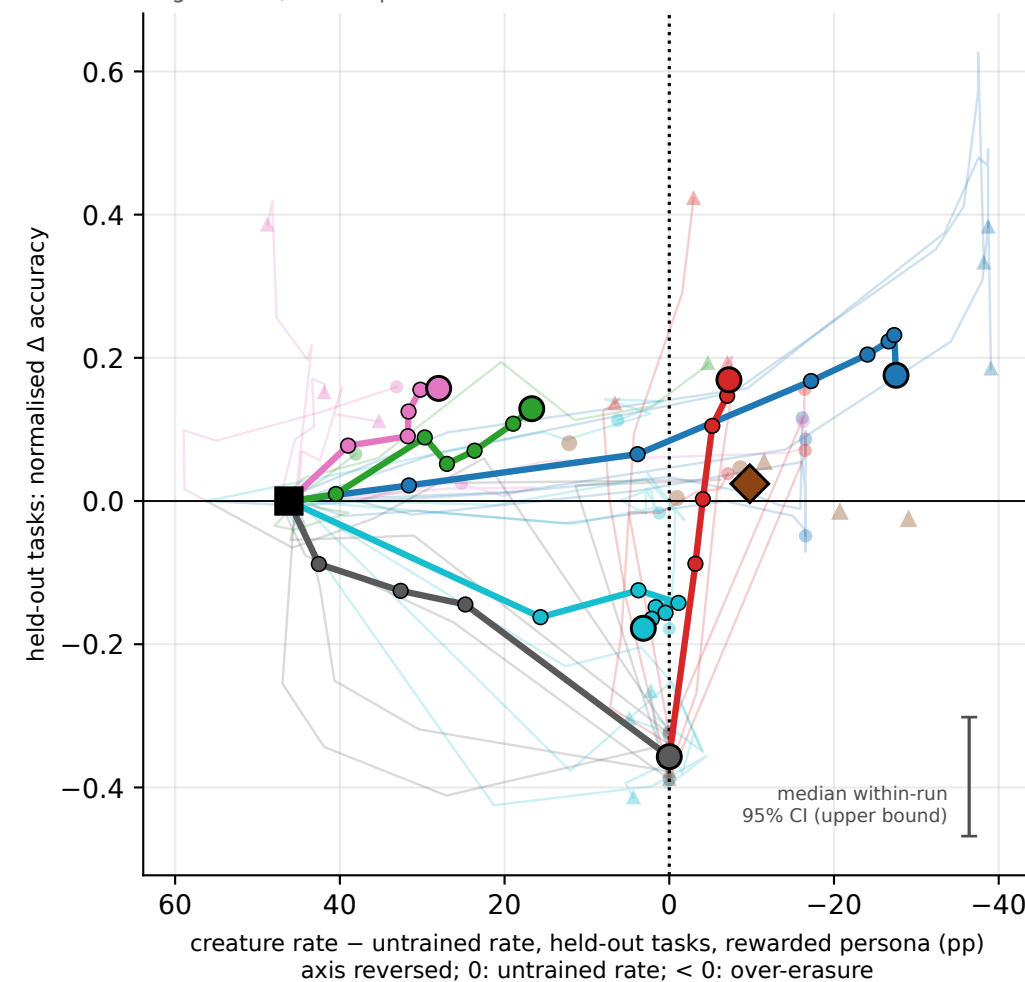
Faint: individual runs. Solid: mean over runs at matched dose; large marker: largest dose; black square: mean hacked anchor.



B. Held-out tasks

Optimal: $x = 0$ with maximal y

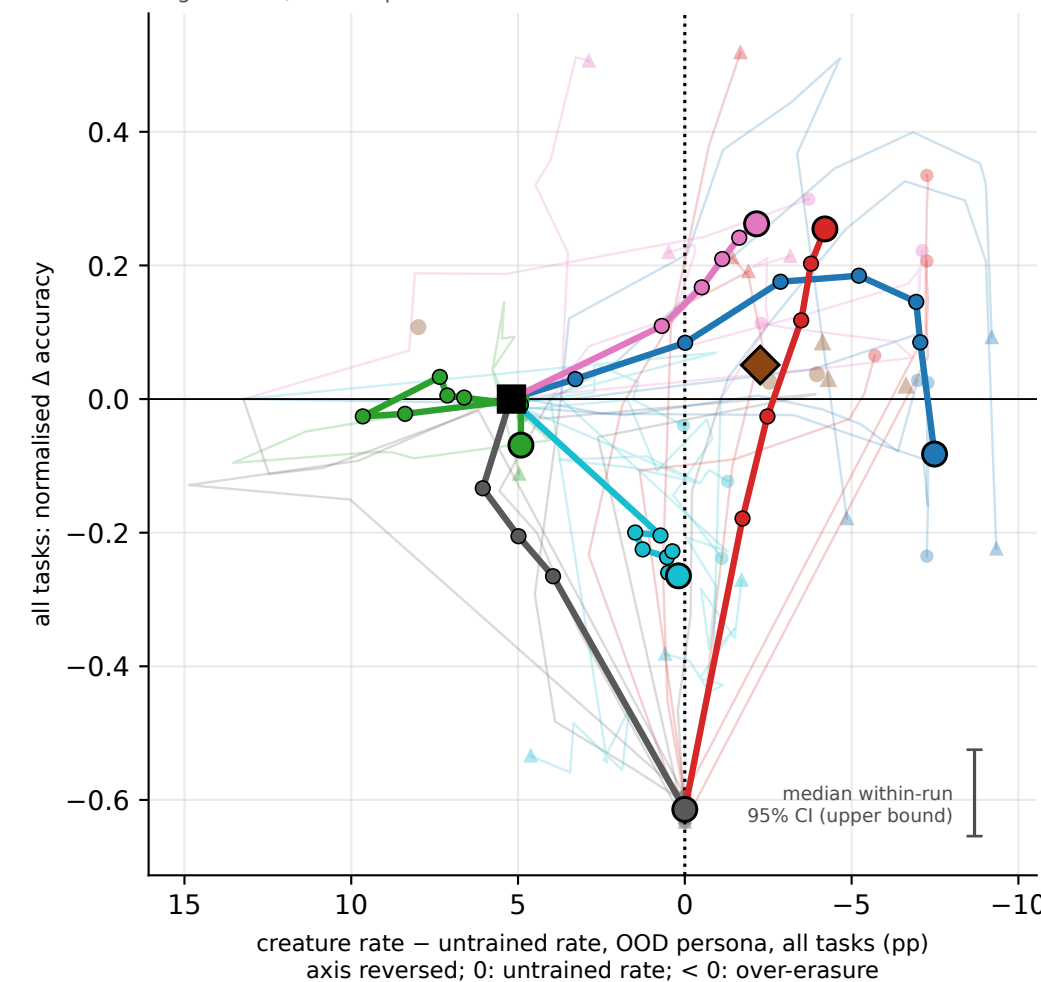
Faint: individual runs. Solid: mean over runs at matched dose; large marker: largest dose; black square: mean hacked anchor.



C. Unrewarded persona

Optimal: $x = 0$ with maximal y

Faint: individual runs. Solid: mean over runs at matched dose; large marker: largest dose; black square: mean hacked anchor.



- Rewind dose: fraction of training undone, interpolated to quarters
- reverse (6 runs)
 - corrected-reward (2 runs)
 - rewind to a checkpoint (6 runs)
 - bc (all prompts, correct) (6 runs)
 - retrain, clean reward (6 runs)
 - continue training, clean reward (6 runs)
 - suppression prompt (6 runs)
 - individual run, Qwen
 - individual run, Gemma
 - hacked anchor